

on monthly earnings, hours worked, and occupations through December 2024—two years after the launch of ChatGPT. Using a difference-in-differences framework, we compare adopters and non-adopters before and after the arrival of AI chatbots, estimating impacts at both the worker and workplace levels.

We first analyze workers who use AI chatbots. If chatbot use enhances individual productivity, a natural question is whether and how such gains affect their earnings (Krueger, 1993). The answer is theoretically ambiguous, depending on whether adoption is worker- or firm-driven (Becker, 1964), whether it displaces users from higher-value tasks (Autor and Thompson, 2025), whether chatbot use is valued as a nonpecuniary amenity (Rosen, 1986), and on the structure of wage setting (Card et al., 2018).

We find precisely null effects of AI chatbot use on earnings and hours. Difference-in-differences estimates for earnings, recorded hours, and wages all center on zero, with confidence intervals ruling out average effects larger than 2%. These null results hold even among workers who use chatbots daily, report substantial productivity gains, perform new AI-related tasks, or work in firms that encourage and invest in chatbots. Moreover, we find no significant effects in any of our 11 occupations, including those with flexible, decentralized wage-setting.

While chatbots have not led to earnings gains for users on average, we find a clear link to occupational mobility: since the arrival of AI chatbots, adopters are significantly more likely to have switched occupations, working about 4% of a full-time equivalent more in their most recent occupation by December 2024. These transitions are most pronounced into IT support specialist and clerical (office support) roles—occupations in which workers have greater discretion over their toolsets and that do not require formal educational credentials. These patterns align with experimental evidence that chatbots are especially beneficial for workers with less prior expertise (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023) and support the view that generative AI helps workers move into new occupations by granting access to otherwise scarce expertise (Autor, 2024). In particular, adopters who switch occupations tend to move into roles with higher wage premia and

greater AI chatbot relevance. On average, these workers see earnings grow 12 percentage points faster than other employed workers. While these switchers are still too few to drive average adopter earnings, this pattern suggests that occupational mobility may be a channel through which AI chatbots deliver broader earnings effects over time (Kleven et al., 2025).

Next, we zoom out to the workplace level and examine whether employers that encourage and invest in AI chatbots have experienced differential changes in labor demand. If chatbots enhance labor productivity, firms may expand hiring; if they substitute for labor, they may slow hiring or trigger layoffs (Acemoglu and Restrepo, 2019). Recent declines in early-career employment in AI-exposed occupations have raised concerns that generative AI is already displacing entry-level jobs (Brynjolfsson, Chandar and Chen, 2025). Crucially, our data allow us to split these employment trends by whether firms have actually adopted generative AI. While we replicate the pattern of declining early-career employment in exposed occupations in Denmark, our difference-in-differences analysis reveals that the declines are not driven by firms adopting AI chatbots. Our estimates are sufficiently precise to rule out even modest changes. Workplaces that encourage chatbot use exhibit no differential changes in employment or wage bills, job creation or destruction, or the composition of hires or separations, including among early-career workers. These null results hold across occupations and persist even when encouragement is paired with enterprise chatbot solutions and training.

Taken together, our findings show that early labor market transformation under generative AI is occurring beneath the surface. Even as wages and recorded hours remain stable, workplaces adopt chatbot initiatives, workers take on new AI-related tasks, and adopters transition into new occupations.

Contributions to the Literature. This paper makes two linked contributions. The first is conceptual: we show how labor markets adjust at the onset of major technological change, arguing that substantial reorganization of work absorbs change before it surfaces in economic statistics. The second is measurement: we develop a unique data infrastructure

linking large-scale, representative data on AI chatbot adoption and work reorganization to administrative labor market records.

A contemporaneous set of studies uses exposure measures—technical assessments of which job tasks are feasible to assist with AI chatbots—to examine whether exposed occupations have fared differently after the release of ChatGPT. Brynjolfsson, Chandar and Chen (2025) document a sharp decline in employment of early-career workers in AI-exposed occupations in the United States since 2022, suggesting that generative AI may already be displacing entry-level jobs. Our key contribution is to split these trends by whether firms have actually adopted generative AI, revealing that the declines are not driven by firms adopting AI chatbots.

A small but growing number of studies field worker-level surveys on the adoption of generative AI (Bick, Blandin and Deming, 2025; Hartley et al., 2025; Humlum and Vestergaard, 2025). While our adoption patterns and self-reported time savings align with these studies, we link these survey measures to administrative labor market outcomes, including earnings and hours at the worker and workplace levels, allowing us to move the analysis beyond self-reported effects.³

While several experiments document sizable productivity gains from chatbot use on specific tasks (Brynjolfsson, Li and Raymond, 2025; Dillon et al., 2025; Noy and Zhang, 2023), it remains unclear how these gains translate into workers’ labor market outcomes, as high-quality microdata are rarely available. While AI chatbot use has not measurably affected workers’ earnings and hours, we find a clear association between adoption and occupational mobility. This aligns with experimental evidence that chatbots provide greater benefits to workers with less prior expertise. Moreover, adopters who switch move into occupations with higher wage premia and where using AI chatbots is more relevant, suggesting that chatbots help workers access otherwise scarce expertise (Autor, 2024; Autor and Thompson, 2025).

³AI labs have recently released descriptive statistics from their internal usage data (Chatterji et al., 2025; Handa et al., 2025; Tomlinson et al., 2025). Our survey data complement these by capturing the full range of chatbot products, measuring complementary workplace investments, linking to administrative records, and sampling both users and non-users from a well-defined population frame.

We provide the first representative account of employer adoption of AI chatbots. Whereas surveys on traditional “pre-generative” AI show relatively low uptake (Acemoglu et al., 2022; Bonney et al., 2024), we find that most employers in exposed occupations have now embraced AI chatbots. Our findings align with Yotzov et al. (2026), who document widespread firm-level adoption of generative AI but limited perceived impacts on employment or revenue productivity. Alternative firm-level data proxy adoption through generative AI skills in job postings (Lichtinger and Hosseini Maasoum, 2025; Schubert, 2025), but it is unclear how well these capture actual tool use, especially for non-specialized tools like chatbots. We address this gap by surveying workers on their chatbot use and employer initiatives, linking responses to administrative firm data. Our survey also allows us to measure complementary workplace investments that otherwise would go unrecorded (Brynjolfsson, Rock and Syverson, 2021).

While employer chatbot initiatives have not led to net changes in earnings or hours, they strongly predict the emergence of new AI-related job tasks (Acemoglu and Restrepo, 2018*a*; Autor et al., 2024). This points to the mechanism behind the still-waters pattern: employers are absorbing generative AI through task reorganization rather than adjusting on the margin of total labor input. This is consistent with recent work on “job transformation,” which emphasizes that new technologies alter the task content of jobs well before they displace them (Autor and Thompson, 2025; Freund and Mann, 2025).

Finally, our account of “still waters over rapid currents” echoes Robert Solow’s famous observation about the IT revolution: “You can see the computer age everywhere but in the productivity statistics” (Solow, 1987)—and is consistent with the “productivity J-curves” observed at the onset of general-purpose technologies (Brynjolfsson, Rock and Syverson, 2021; McElheran et al., 2025). We document the hallmarks of such a trough: widespread adoption accompanied by substantial employer investments in training, integration, and oversight that have yet to translate into hard economic outcomes. Our unique data infrastructure reveals what lies beneath the still waters: the reorganization of work that absorbs technological change before it surfaces in hours or earnings.